

Exercise 1 (Coefficient of determination):

The estimated model in Exercise 1 on Sheet 1 resulted in an R^2 (Coefficient of Determination) of 0.038. What is the correlation between an individual's wage and his age? Interpret your results and provide one possible explanation.

Exercise 2:

Consider the following model:

$$\begin{aligned} wage = & \beta_0 + \beta_1 \cdot age + \beta_2 \cdot HSGrad + \beta_3 \cdot SomeCollege + \beta_4 \cdot CollegeGrad \\ & + \beta_5 \cdot AdvancedDegree + \beta_6 \cdot health.Verygood + \varepsilon. \end{aligned}$$

The method of least squares resulted in the following estimation:

$$\hat{\beta}'_{LS} = (52.61, 0.62, 11.01, 23.16, 37.97, 62.63, 9.13)$$

- Assuming normally distributed errors, what is the Maximum Likelihood estimation for β ?
- Calculate the Restricted Maximum Likelihood estimation for the model variance σ^2 . Use $\hat{\varepsilon}'\hat{\varepsilon} = 3819720$.
- John is a 37 year old man with education level 4 (College Grad) and very good health. Calculate his estimated wage.
- John's real wage is \$121.43. What is his residual for the model above? Interpret this residual with one sentence. What is his standardized residual? (Hint: for the latter part you can use: $h_{ii} = 0.0016$)
- Calculate the estimated standard errors of the regression coefficients. Use

$$(\mathbf{X}'\mathbf{X})^{-1} = 10^{-5} \cdot \begin{pmatrix} 939.15 & -11.46 & -361.78 & -366.68 & -333.97 & -301.47 & -146.10 \\ -11.46 & 0.26 & -0.16 & 0.12 & -0.45 & -1.08 & 1.03 \\ -361.78 & -0.16 & 476.55 & 373.93 & 374.84 & 375.57 & -8.13 \\ -366.68 & 0.12 & 373.93 & 529.27 & 376.65 & 377.28 & -18.95 \\ -333.97 & -0.45 & 374.84 & 376.65 & 526.07 & 382.70 & -34.01 \\ -301.47 & -1.08 & 375.57 & 377.28 & 382.70 & 621.96 & -44.49 \\ -146.10 & 1.03 & -8.13 & -18.95 & -34.01 & -44.49 & 172.52 \end{pmatrix}$$

Hint: You do not need the whole matrix.